

- 6 An electric guitar uses electromagnetic pickups to detect the vibration of its strings, which are made from steel. Each pickup consists of a coil of wire wrapped around a permanent magnet as shown in Fig. 6.1. When the string vibrates near the pickup, an alternating electromotive force (e.m.f.) is induced in the coil (which is channelled to an output amplifier and speaker)

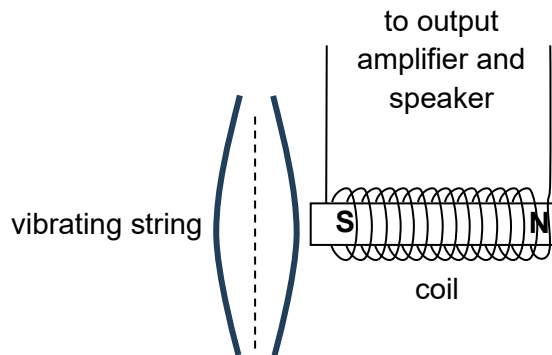


Fig. 6.1

- (a) Explain how the vibration of the steel string leads to the generation of an alternating e.m.f. in the coil of the pickup.

The steel string is made of magnetic material, and so it causes changes in the magnetic field from the permanent magnet that the coil is situated in.

As the string vibrates, the magnetic field near the (pickup) coil changes over time. [B1]

A changing (periodically increasing and decreasing) magnetic flux linkage through the coil [B1] induces an electromotive force in the coil proportional to the rate of change of magnetic flux [B1]

Because the vibration is periodic, the magnetic flux linkage also changes periodically and thus the e.m.f. induced is alternating [B1]

.....[4]

- (b) A guitarist plucks a string more strongly, causing it to vibrate with the same frequency but larger amplitude. Explain how this affects the e.m.f. induced in the coil.

Larger amplitude of vibration with the same frequency causes greater change in magnetic field around the magnet and coil, so there will be a greater change in magnetic flux linkage in coil in the same period, hence greater magnitude of induced voltage.

.....[2]

- (c) A guitar string vibrates in a magnetic field, where the field strength at the coil fluctuates between maximum and minimum with a difference of field strength of $5.0 \times 10^{-2} \text{ T}$ at a frequency of 880 Hz. The coil has a cross-sectional area of $1.0 \times 10^{-6} \text{ m}^2$. It is desired that the average e.m.f. of the pickup coil is about 0.20 V.

Estimate the number of turns the coil must have.

$$|E| = \frac{d\Phi}{dt} = NA \frac{dB}{dt} = NA \frac{\Delta B}{\Delta t} \quad \frac{\Delta B}{\Delta t} = \frac{\Delta B_{\text{max to min}}}{T/2} = 2(\Delta B)f$$

$$N = \frac{|E|}{A \left(\frac{\Delta B}{\Delta t} \right)} = \frac{0.20}{(1.0 \times 10^{-6})(5.0 \times 10^{-2})(2 \times 880)} \quad [M1]$$

$$= 2272 \text{ turns} \approx 2 \times 10^3 \quad [A1]$$

number of turns = [2]

[Total: 8]

